OSTEOPATHIC APPROACH TO THE HOSPITALIZED PATIENT

Stacia Sloane, DO

With thanks to Dr. Vanessa Newman and Dr. Melissa Pearce

OSTEOPATHIC NEUROMUSCULOSKELETAL MEDICINE (ONMM)

- Inpatient consult service: trauma, surgery, medicine, neonates, pediatrics, post-partum
- Outpatient clinic: neonates, peds, expecting/postpartum mothers, adults, geriatrics
- Longitudinal clinics in rheum, sports med, ortho, neuro, radiology

OBJECTIVES

1. Recognize the range of benefits of using OMM in the treatment of hospitalized patients
2. Recognize the five models of osteopathic medicine
3. Comprehend the importance of optimizing lymphatic and respiratory function in wide range of acutely ill patients in context of respiratory –circulatory model
4. Understand viscerosomatic reflexes and recognize their clinical importance
5. Strategize on how best to discuss OMM as a medical student when on clinical rotations with attendings, care team, patients and family
6. Comprehend components of osteopathic treatment plan/ prescription in hospitalized patient
7. Recognize good candidates for inpatient OMM and understand why
8. Familiarize with the logistics of inpatient OMM
9. Recognize possible post-operative complications, esp. atelectasis/pneumonia, ilius, wound infection, hospital length of stay/readmission
10. Apply appropriate OMT to avoid and/or treat post-operative complications
11. Recognize pertinent study outcomes in hospitalized and post-surgical patient populations
12. Optimize opportunities for utilizing OMT in pre- and post-surgical patient visits
13. Recognize red flags and contraindications in hospitalized and surgical patients

OUTLINE- INPATIENT OMT

- Osteopathic philosophy: What sets us apart as osteopathic physicians?
- Who can we treat?
- How do we treat?
  - Models of osteopathy/ABCs
  - Treatment plan/dosing
  - Logistics
  - Osteopathic structural exam inpatient
- Communication
- Case #1: GSW to leg
- Respiratory- circulatory considerations:
  - Lymphatics
  - Respiration
- Case #2: Pneumonia
- Viscerosomatic reflexes

OSTEOPATHIC PRINCIPLES:

- The patient is a unit of body, mind and spirit
- Capable of self-regulation, self-healing, health maintenance
- Structure and function are reciprocally interrelated
- Rational treatment is based upon the above

OSTEOPATHIC MANIPULATIVE TREATMENT

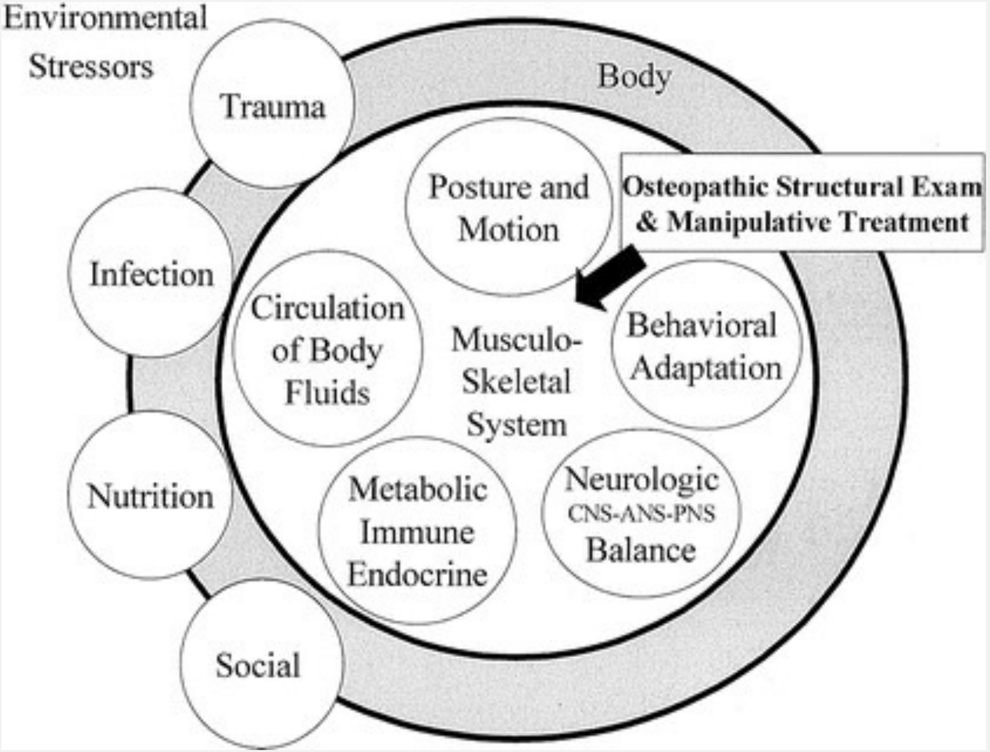
- Applicable to **full spectrum of medical and surgical** problems
- Based on understanding **mechanical and functional** aspects of the body's viscera and systems
- Addresses underlying **physiologic** mechanisms of the disease and supports the **physiologic response** of the host who has the disease
    - Immune response, oxygenation, perfusion, lymphatic flow
- Improve patient satisfaction/comfort
- Prevent/treat common complications of hospitalization and surgery

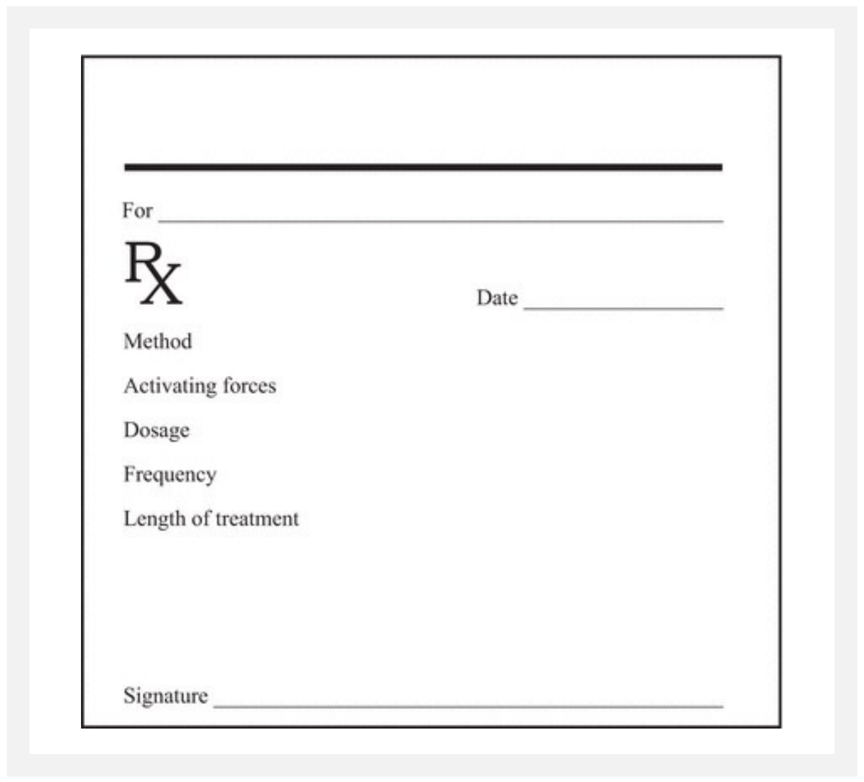
WHO CAN WE TREAT INPATIENT WITH OMM?

- Medicine
- Surgical
- Ortho
- Trauma
- OBGYN
- Pediatrics

HOW DO WE TREAT INPATIENT?

- Five models of osteopathic care ** board relevant!**
  - Biomechanical Model
  - Respiratory-circulatory Model
  - Neurological Model
  - Metabolic energy Model
  - Behavioral/ Biopsychosocial Model
- ABCs
  - Autonomics, Biomechanics, Circulation, screens




For
Rx
Date
Method
Activating forces
Dosage
Frequency
Length of treatment
Signature

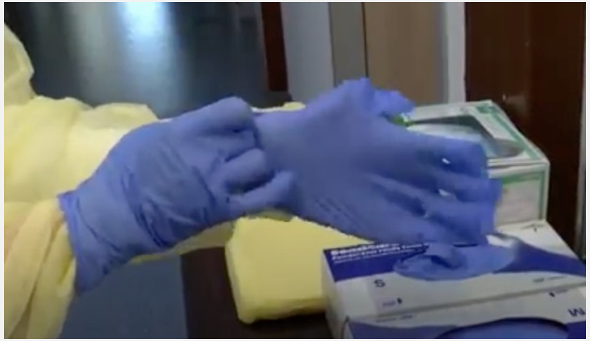
HOW DO WE TREAT INPATIENT? **TREATMENT PLAN**!:

- Which areas most impair the patient's homeostatic balance
- Which techniques are best suited for the patient
- How much treatment the patient can tolerate

HOW DO WE DECIDE **WHERE** ON OUR PATIENT'S BODY TO TREAT?

- Osteopathic structural exam!
  - Start at the feet, work your way up the pt's body toward their head
  - Fluid wave, leg traction, fascial pull
  - Particular attention to the junctions (LS, TL, CT, OA)--> diaphragms
    - Zink screen
  - Rib motion
  - TART changes in thoracic/upper lumbar spine
    - Suggestive of viscero-somatic/somato-somatic reflexes?
  - Chapman's Points

HOW DO WE TREAT?
INPATIENT OMT



- LOGISTICS!
  - How do you get your hand under a supine patient?
    - Turn gloves inside out
    - Use sheets, chucks
  - Gracefully arrange all lines, ventilator, etc. Know what you are touching!
  - Remember ergonomics- adjust bed as needed

DOCUMENTATION

- OMT is recorded with a procedure note for purposes of billing and documentation
- Procedure notes are often built into the EMR at the hospital or clinic where OMM happens
- When you do some OMT as part of a larger visit you can add relevant details to your H&P or SOAP note
- Record elements of your structural exam, the treatments used, the patient response
- List OMT is part of the overall plan for that inpatient visit
- Anything you do requires PERMISSION FROM YOUR ATTENDING

COMPREHENSIVE EXAM WORKSHEET

- Example of Osteopathic Structural Exam table
- Brief Plan

OMM Worksheet

Student Doctor Name:__________________________ **Partner Name:**____________________

Chief complaint:____

Standing/ Static Exam general observation:		**Gait/posture:**				
Musculoskeletal **(TART)** FINDINGS		DNE*	NSF*	Region Evaluated	Osteopathic Structural Exam **(OSE)**	Worst SD
		☐	☐	Head **(Cr/OA)**		
		☐	☐	Cervical **(AA/C)**		
		☐	☐	Thoracic **(T)**		
		☐	☐	Ribs **(R)**		
		☐	☐	Lumbar **(L)**		
		☐	☐	Innominates/ Pelvis **(P)**		
		☐	☐	Sacrum **(S)**		
		☐	☐	Lower Extremity **(LE)**		
		☐	☐	Upper Extremity **(UE)**		
		☐	☐	Abdomen/ Other **(ABD)**		

*DNE= Did Not Examine /*NSF= No Significant Findings **You must have findings & diagnosis, or one of the circles checked for each region.**

A: Autonomics:

B: Biomechanics:

C: Circulation:

S: Screening:

Name: Date of Encounter:
Date of Birth: Age:

TOURO UNIVERSITY CALIFORNIA

Student Doctor's Note

This form is for student educational purposes only, and is not intended to constitute medical documentation

CC:

HPI:

ROS: ▢No new signs/symptoms / ▢New signs/symptoms *v = asked, x = new signs/ symptoms*
▢Const. ▢HEENT ▢CV ▢Resp. ▢GI ▢GU ▢Neuro ▢MSK ▢Skin ▢Endo ▢Heme/Lymph ▢Allergy/Imm ▢Psych

Family/Social History: ▢No new relevant data ▢New data:
Medications: ▢No new meds ▢New meds (→update summary sheet)
Allergies: ▢No new allergies ▢New allergies (→update summary sheet)

Physical exam:

Vitals: BP___/___ HR ___ beats/min RR ___ breaths/min Temp ___ ○F/○C Weight ___ lbs/kg Height _____ in/cm
General:
Mental Status: Mood / Affect: ▢situationally appropriate ▢depressed ▢flat ▢anxious ▢agitated ▢in pain
Posture:

Gait: ▢Symmetrical & balanced ▢Asymmetrical & uneven ▢Antalgic ▢Other:___________________________

Organ systems: ▢HEENT ▢Neck ▢CV ▢Resp. ▢Abdominal ▢GU/Breast ▢Skin *v= examined x= new signs/symptoms*

Ortho/Neuro:

Sensation **Strength** **DTR's** **Special Tests:**



Peds: Head circumference: __________ in/cm Fontanelles: open / closed / flat / bulging / sunken
Peds Reflexes: ▢ Moro ▢ Suck ▢ Rooting ▢ ATNR ▢ Galant ▢ Palmar grasp ▢ Plantar grasp ▢ Babkinski

Screen	Region	Somatic Dysfunctions and Other Symptoms	Severity 1= mild 2= moderate 3= severe	Treatment Method
Gravitational Line:	Head & OA			ST ART LAR ME OCF MFR BLT FPR SCS VIS HVLA FDM BD DIR IND
	Cervical			ST ART LAR ME OCF MFR BLT FPR SCS VIS HVLA FDM BD DIR IND
	Thoracic T 1-4			ST ART LAR ME OCF MFR BLT FPR SCS VIS HVLA FDM BD DIR IND
	T 5-9			ST ART LAR ME OCF MFR BLT FPR SCS VIS HVLA FDM BD DIR IND
Zink: LA___ CC___ CT___ TL___ LS___ LL ___	T 10-12			ST ART LAR ME OCF MFR BLT FPR SCS VIS HVLA FDM BD DIR IND
	Ribs			ST ART LAR ME OCF MFR BLT FPR SCS VIS HVLA FDM BD DIR IND
	Lumbar			ST ART LAR ME OCF MFR BLT FPR SCS VIS HVLA FDM BD DIR IND
	Pelvis/ Innominate			ST ART LAR ME OCF MFR BLT FPR SCS VIS HVLA FDM BD DIR IND
Fluid Wave:	Pelvis/ Sacrum			ST ART LAR ME OCF MFR BLT FPR SCS VIS HVLA FDM BD DIR IND
	Upper Extremity			ST ART LAR ME OCF MFR BLT FPR SCS VIS HVLA FDM BD DIR IND
	Lower Extremity			ST ART LAR ME OCF MFR BLT FPR SCS VIS HVLA FDM BD DIR IND
	Abdomen/ Other			ST ART LAR ME OCF MFR BLT FPR SCS VIS HVLA FDM BD DIR IND

Diagnoses:
1.__________________________ 2. __________________________ 3. __________________________
4. Somatic Dysfunction(s):
▢ Head/OA ▢ Cervical ▢ Thoracic ▢ Lumbar ▢ Ribs ▢ Pelvis/Innominate ▢ Pelvis/Sacrum
▢ Upper Extremity ▢ Lower Extremity ▢ Abdomen/Other:__________________

Treatment Plan:
1. Evaluation prior to treatment: ▢Resolved ▢Improved ▢Unchanged ▢Worse ▢New patient
2. OMT tolerated _______________ treatment comments:
3. Osteopathic manipulative treatment performed as above to:
   ▢ 1-2 regions ▢ 3-4 regions ▢ 5-6 regions ▢ 7-8 regions ▢ 9-10 regions ▢ >10 regions
4. Counseled patient/parent:
   ▢ exercise ▢ modification of activity ▢ lifestyle factors ▢ nutrition ▢ post-treatment discomfort, rest, hydration
   Instructions:
5. Minutes spent with patient: 10 20 30 40 60 >60 minutes
6. Re-evaluate in: _____ Days _____ Weeks _____ Months _____PRN
7. Follow-up notes:

Student Doctor(s):________________________ Signature:______________________________Date:____________

Physician Preceptor:
▢ Glover ▢ Moloney ▢ Sloane ▢ Nuño ▢Pearce Signature:______________________________Date:____________
▢ Peña ▢ Wolf ▢ Thom ▢ Other:

COMMUNICATION IS KEY!

- Define terms:
  - DO- (Student) Doctor of Osteopathy
  - OMM- Osteopathic manipulative medicine
  - OMT- Osteopathic manipulative treatment
- Know your audience: patient, MD attending, DO attending
- Keep it scientific- OMM is based on anatomy and physiology– all medical practitioners can relate to it with gentle guidance.
- To the patient: "Hello, I'm Student Doctor X. I am trained to use my hands to very carefully examine and treat the body. I will very gently look for any obstructions or areas that aren't moving well and I will use gentle techniques to remove these obstructions and get things moving better. Just let me know if anything is uncomfortable and I can adjust as I go.“

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CASE 1: GUNSHOT WOUND (GSW)

- CC: Severe right leg pain.
- HPI: 20 yo M with PMH of stab wounds x 5 to the lower extremities (2017) brought in by EMS as a trauma code s/p GSW to the L buttocks (DOI: 11/9/19), now s/p ORIF R femur fracture (DOS: 11/10/19). OMT consulted on admission by trauma dept to optimize pt's clinical course.
- Pain is moderately controlled with percocet, gabapentin, ibuprofen. Admits to persistent swelling with numbness of right leg to medial aspect of ankle.
- ROS negative except as noted above


PORTABLE
R

PHYSICAL EXAM (PE):

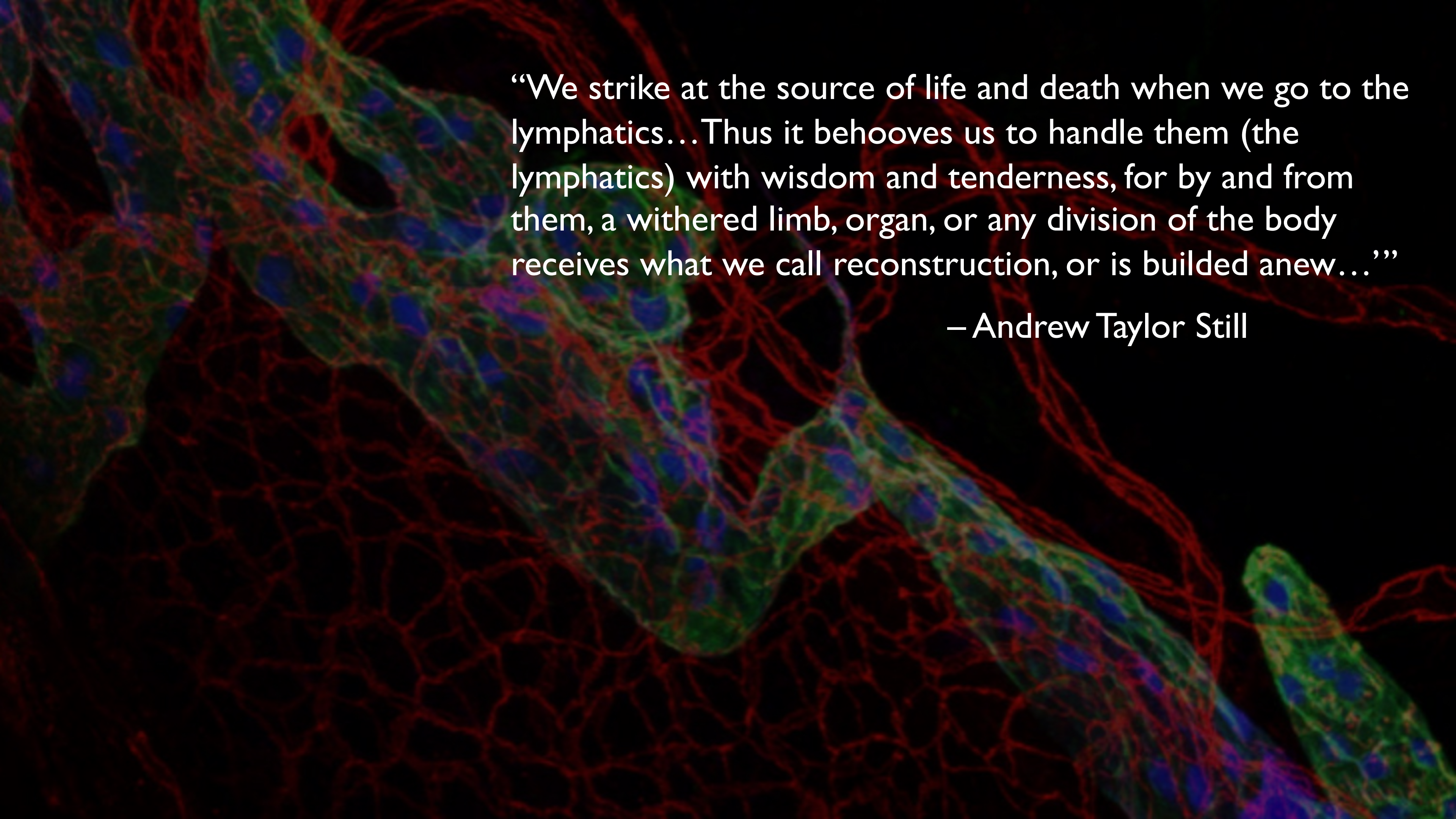
- Vitals: WNL
- Gen: NAD, A&Ox3
- Head/neck: NCAT, supple
- CV: RRR, +S1/S2, no murmur, rubs, gallops, thrills
- Resp: CTA b/l no wheeze, rhonchi, crackles. **Reduced air movement in the bases.**
- Abd: normoactive, soft, NT/ND
- Ext: bandages c/d/i, **non-pitting edema throughout RLE to hip**, soft compressible compartments, peripheral pulses 2+/4 equal. Reported **decreased sensation** to light touch medial aspect of RLE to ankle.

OSTEOPATHIC STRUCTURAL EXAM (OSE):

- Head: OA ESRRL; right condyle compressed
- Cervical: C2 RR; CTJ FRSR
- Thoracic: **Thoracolumbar junction (TLJ) with increased warmth and boggines on the R>L, decreased motion w/ respiration**
- Lumbar: L1-L2 ERSL; L5 F RLSR; LSJ compression
- Sacrum: base posterior; **R SI** resists lateral traction
- Pelvis: **right innominate posterior; right innominate inflared**
- LE: **right hip ER with increased tension through inguinal ligament; decreased PRM**
- UE: **left clavicle adducted w/ mild supraclavicular bogginess**
- Ribs: Left first rib inhaled; decreased excursion and compliance throughout
- Abdomen: **R>L hemidiaphragm inhaled**; **b/l crus restriction**; **R psoas hypertonic**

ASSESSMENT AND PLAN:

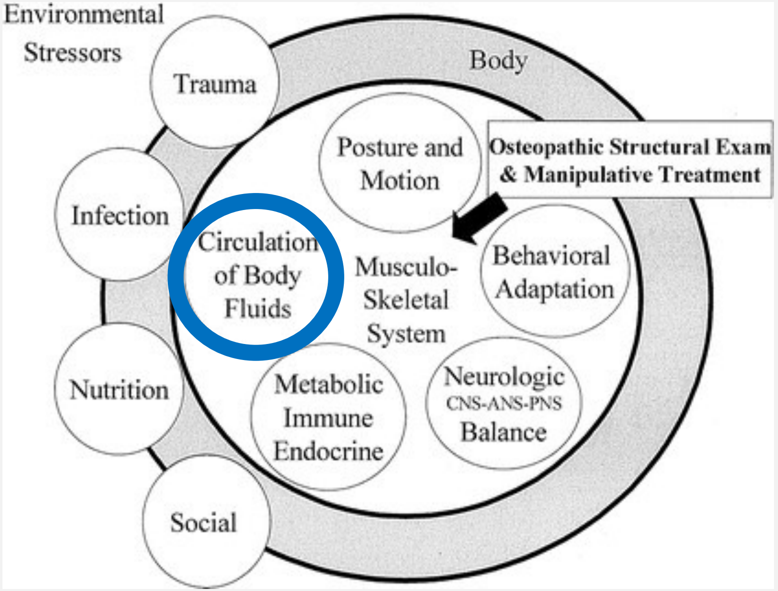
- 20M with PMH noted above, s/p GSW to the L buttocks, now s/p ORIF R femur fracture (DOS: 11/10/19).
- Post op xr w/ good alignment.
- Severe swelling; soft, compressible, NVI.
- OSE w/ above noted severe acute and chronic SD, most notable in RLE, pelvis, sacrum, diaphragm and thoracic inlet, c/w traumatic strain and respiratory-circulatory dysfunction. After discussing risks/benefits/alternatives, consent was obtained, OMT, primarily BLT, was performed to palliate pain, decrease traumatic strain, optimize resp-circ function to promoting peripheral and central lymph drainage and support post-op healing. Treatment was well tolerated with noted improvement in severity of SD.

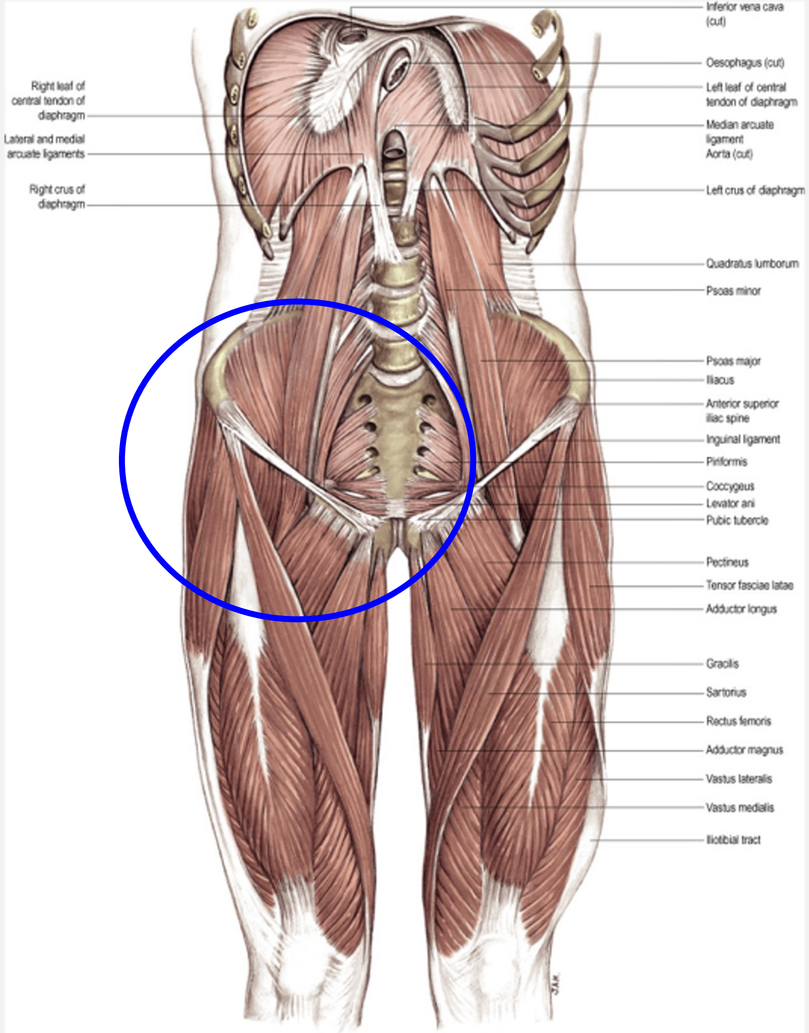

"We strike at the source of life and death when we go to the lymphatics… Thus it behooves us to handle them (the lymphatics) with wisdom and tenderness, for by and from them, a withered limb, organ, or any division of the body receives what we call reconstruction, or is builded anew…'"
– Andrew Taylor Still

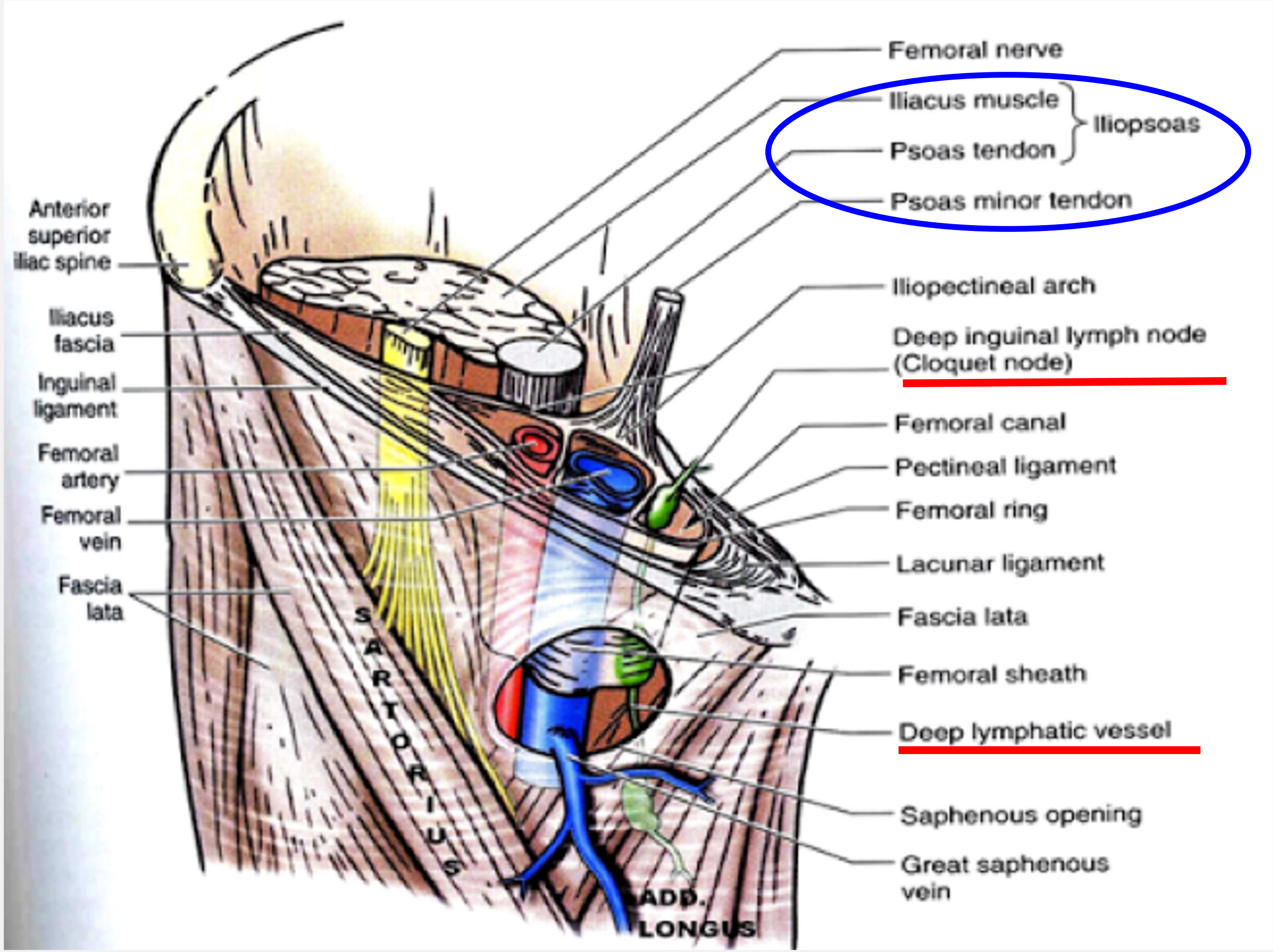
RESPIRATORY-CIRCULATORY CONSIDERATIONS

- Movement of body fluids and critical gas exchange
  - Lymphatics
  - Respiration

- Essential to consider in every patient, especially inpatient!

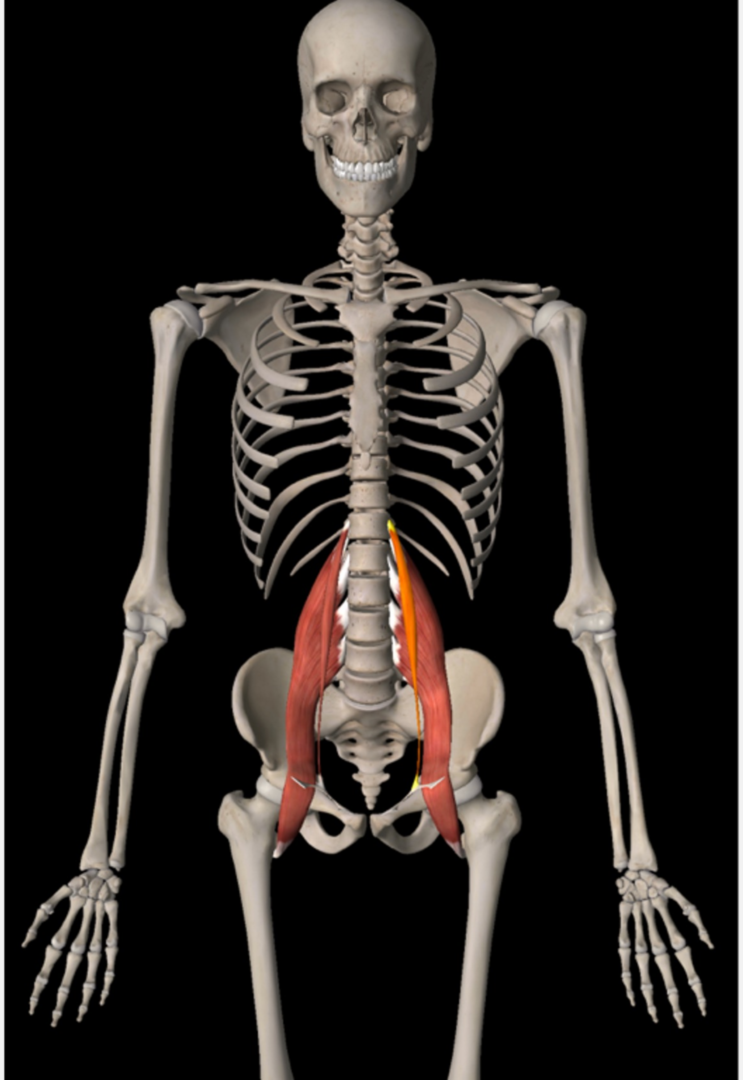


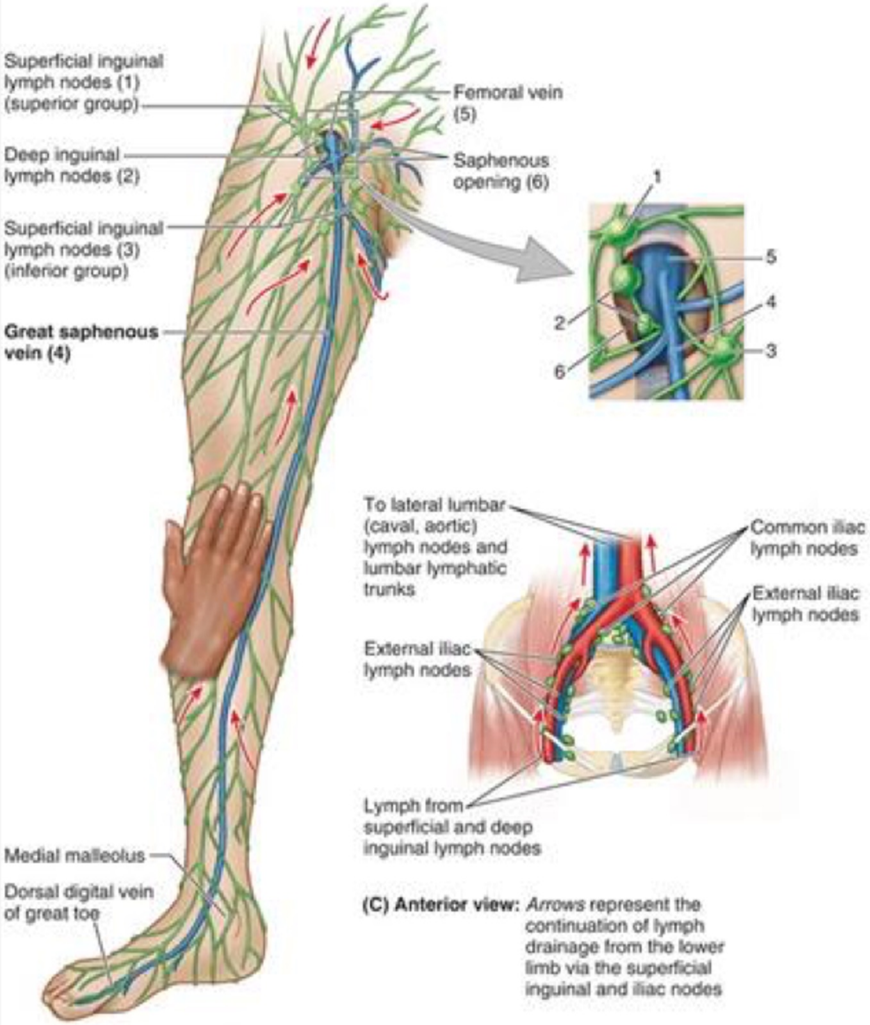

Inferior vena cava
(cut)
Oesophagus (cut)
Left leaf of central
tendon of diaphragm
Median arcuate
ligament
Aorta (cut)
Left crus of diaphragm
Quadratus lumborum
Psoas minor
Psoas major
Iliacus
Anterior superior
iliac spine
Inguinal ligament
Piriformis
Coccygeus
Levator ani
Pubic tubercle
Pectineus
Tensor fasciae latae
Adductor longus
Gracilis
Sartorius
Rectus femoris
Adductor magnus
Vastus lateralis
Vastus medialis
Iliotibial tract
Right leaf of
central tendon of
diaphragm
Lateral and medial
arcuate ligaments
Right crus of
diaphragm



Femoral nerve
Iliacus muscle
Psoas tendon
Iliopsoas
Psoas minor tendon
Iliopectineal arch
Deep inguinal lymph node (Cloquet node)
Femoral canal
Pectineal ligament
Femoral ring
Lacunar ligament
Fascia lata
Femoral sheath
Deep lymphatic vessel
Saphenous opening
Great saphenous vein
Anterior superior iliac spine
Iliacus fascia
Inguinal ligament
Femoral artery
Femoral vein
Fascia lata
SARTORIUS
ADD. LONGUS

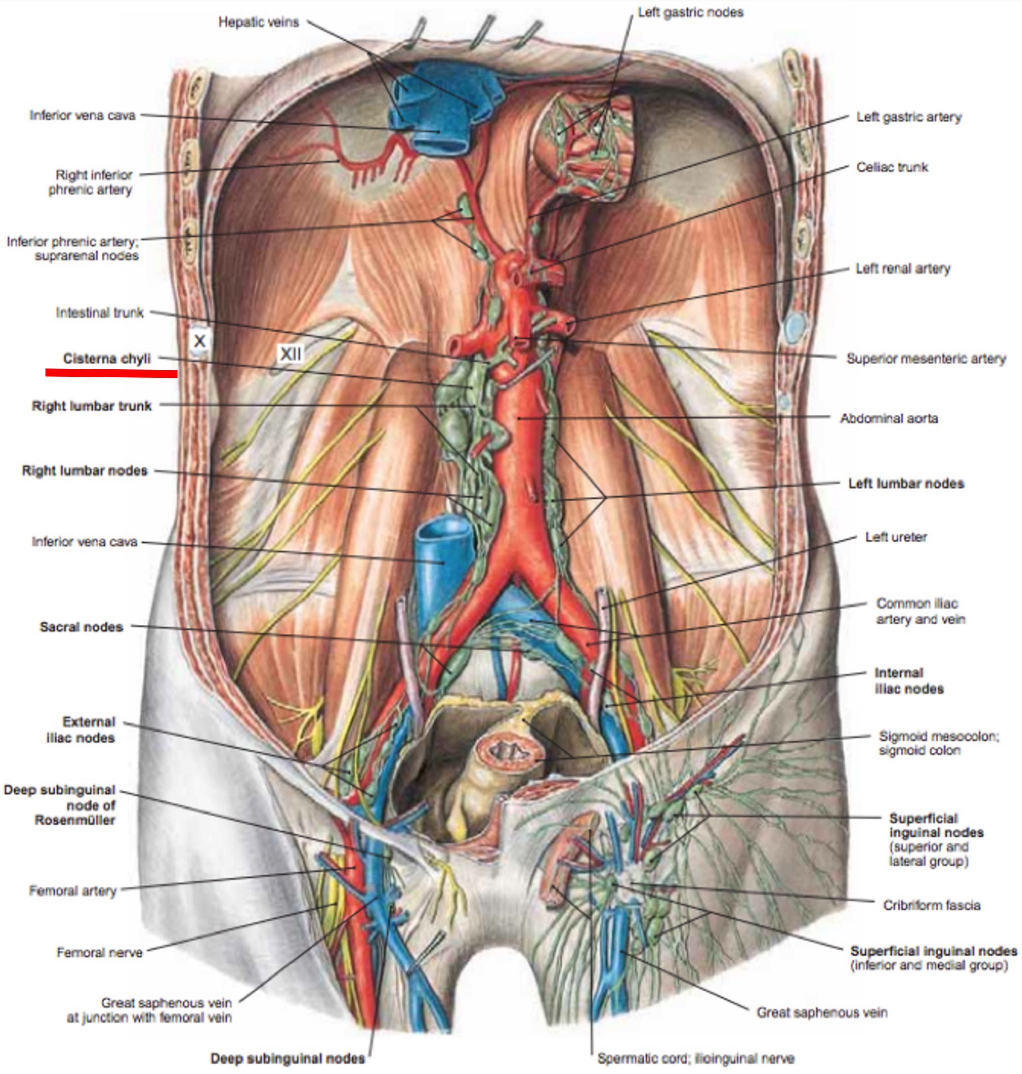
NAVL

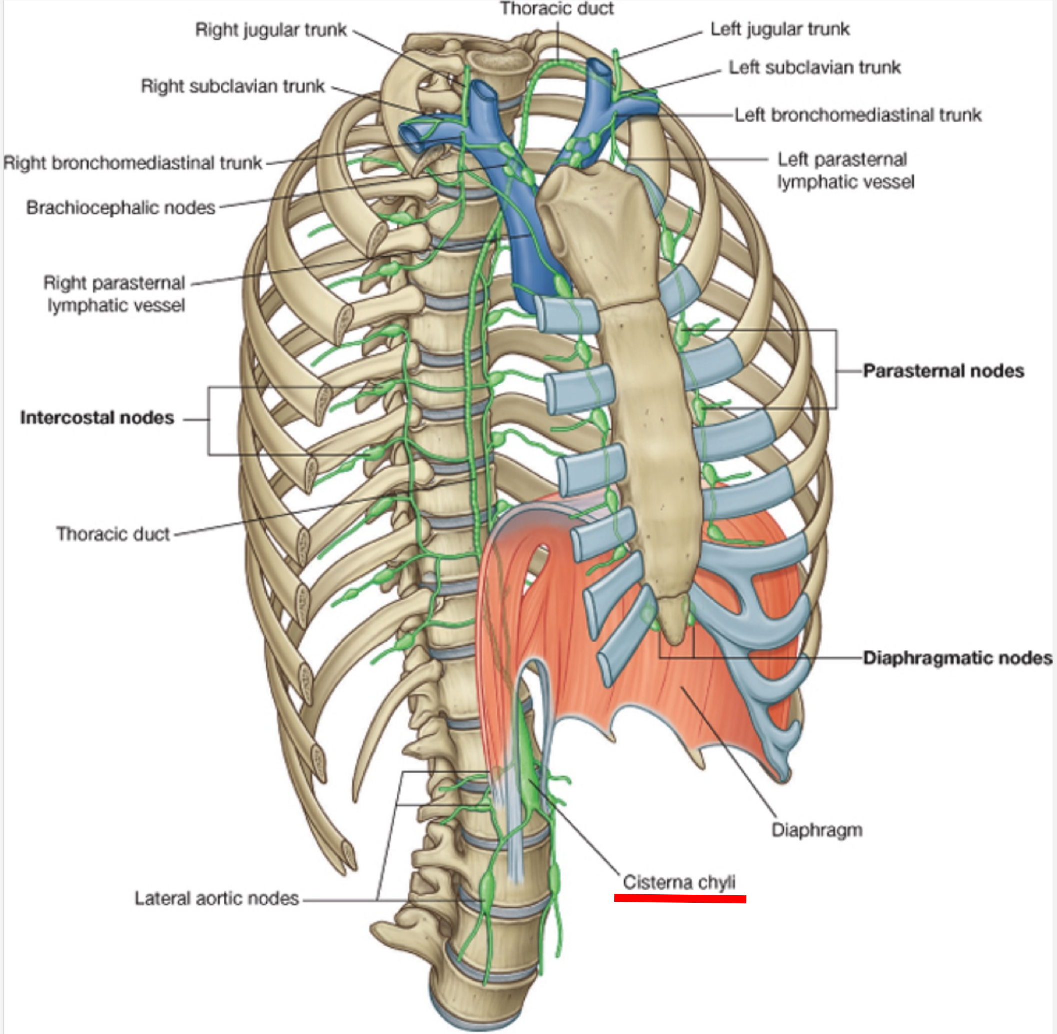




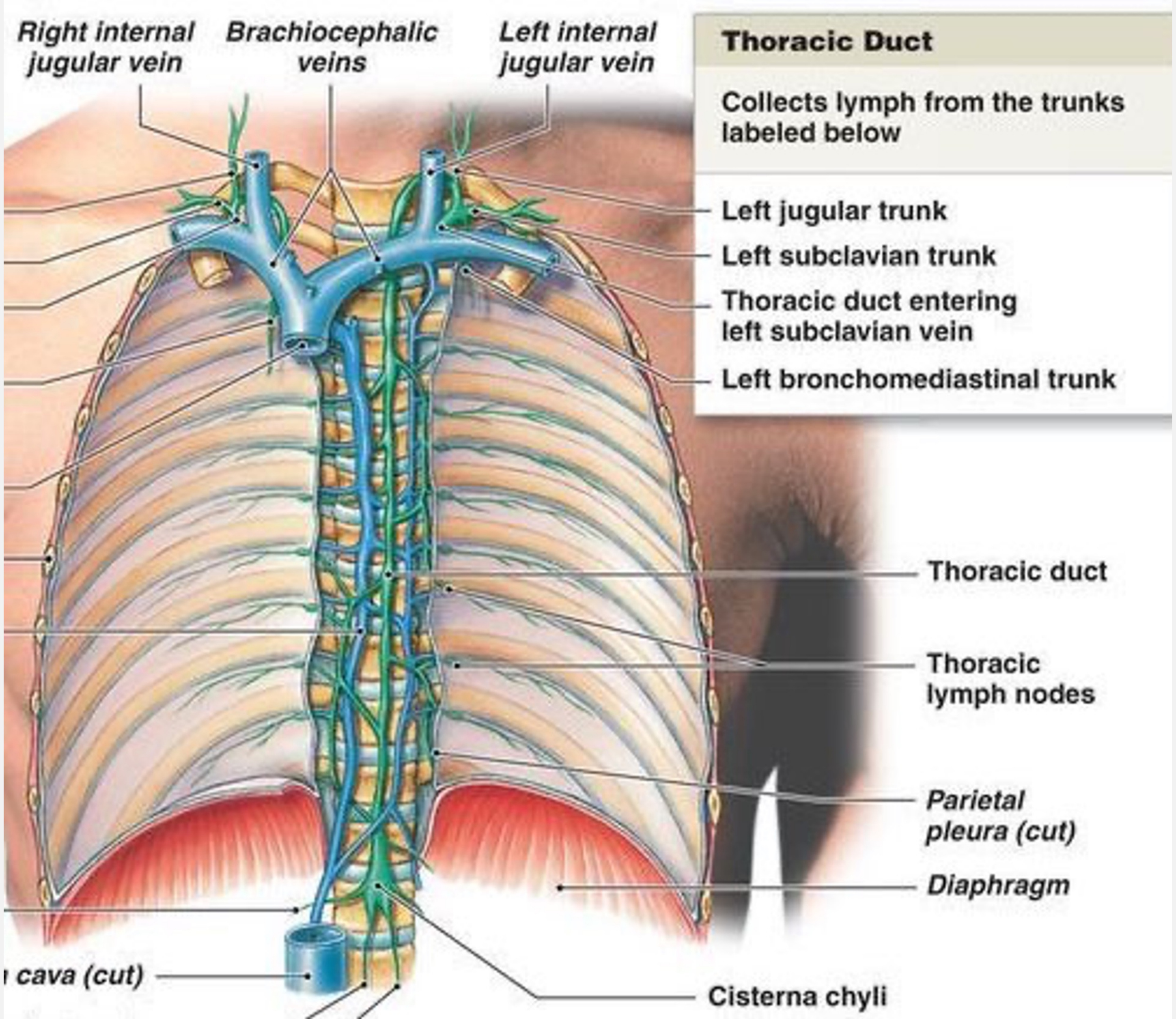
(A) Anteromedial view of right limb

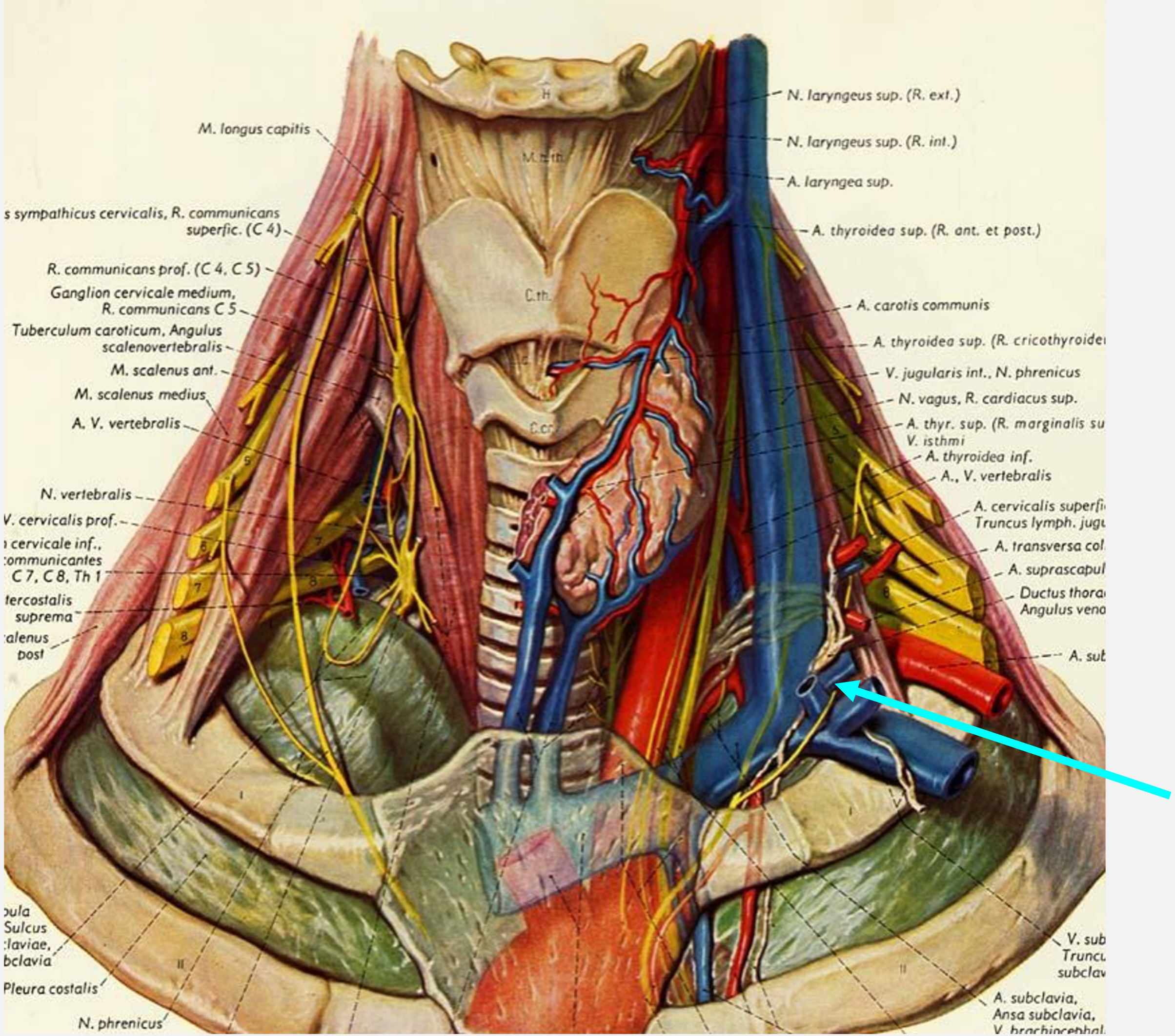
1. Inguinal nodes. 2. External iliac node.


Hepatic veins
Left gastric nodes
Inferior vena cava
Left gastric artery
Celiac trunk
Right inferior phrenic artery
Inferior phrenic artery; suprarenal nodes
Left renal artery
Intestinal trunk
X
XII
Superior mesenteric artery
Cisterna chyli
Right lumbar trunk
Abdominal aorta
Right lumbar nodes
Left lumbar nodes
Inferior vena cava
Left ureter
Common iliac artery and vein
Sacral nodes
Internal iliac nodes
External iliac nodes
Sigmoid mesocolon; sigmoid colon
Deep subinguinal node of Rosenmüller
Superficial inguinal nodes (superior and lateral group)
Femoral artery
Cribriform fascia
Femoral nerve
Superficial inguinal nodes (inferior and medial group)
Great saphenous vein at junction with femoral vein
Great saphenous vein
Deep subinguinal nodes
Spermatic cord; ilioinguinal nerve

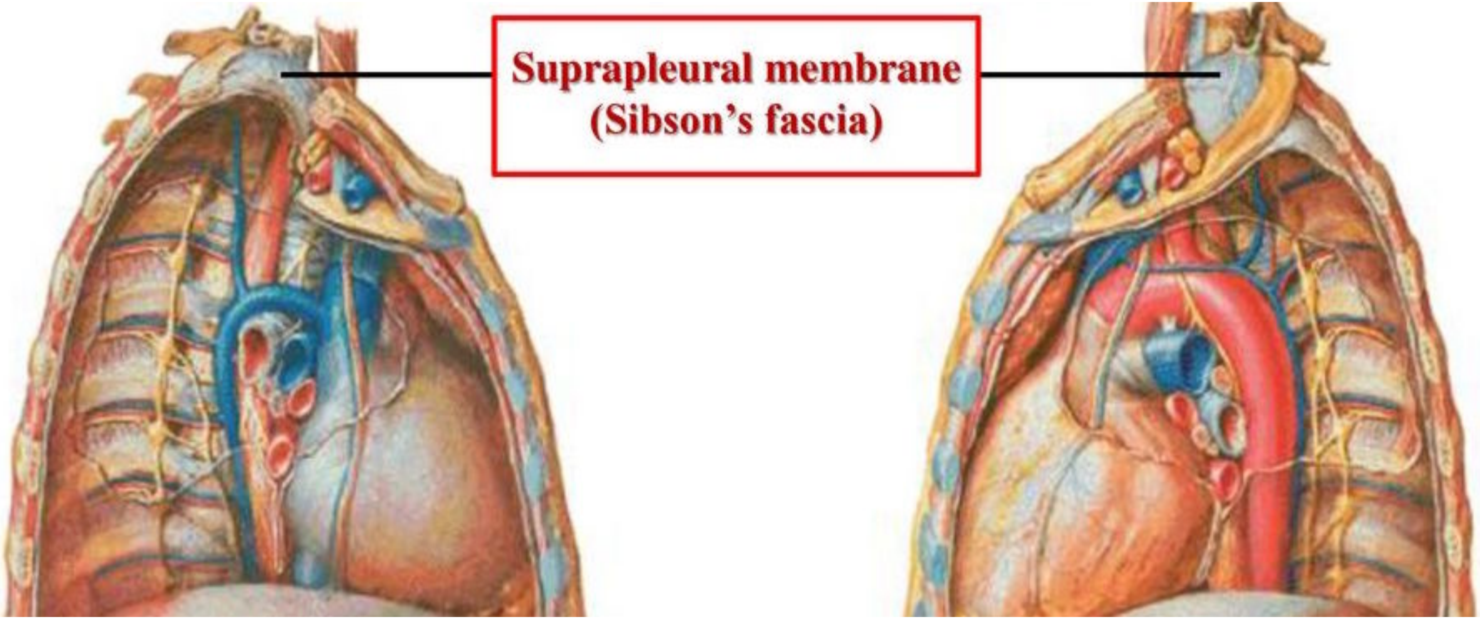


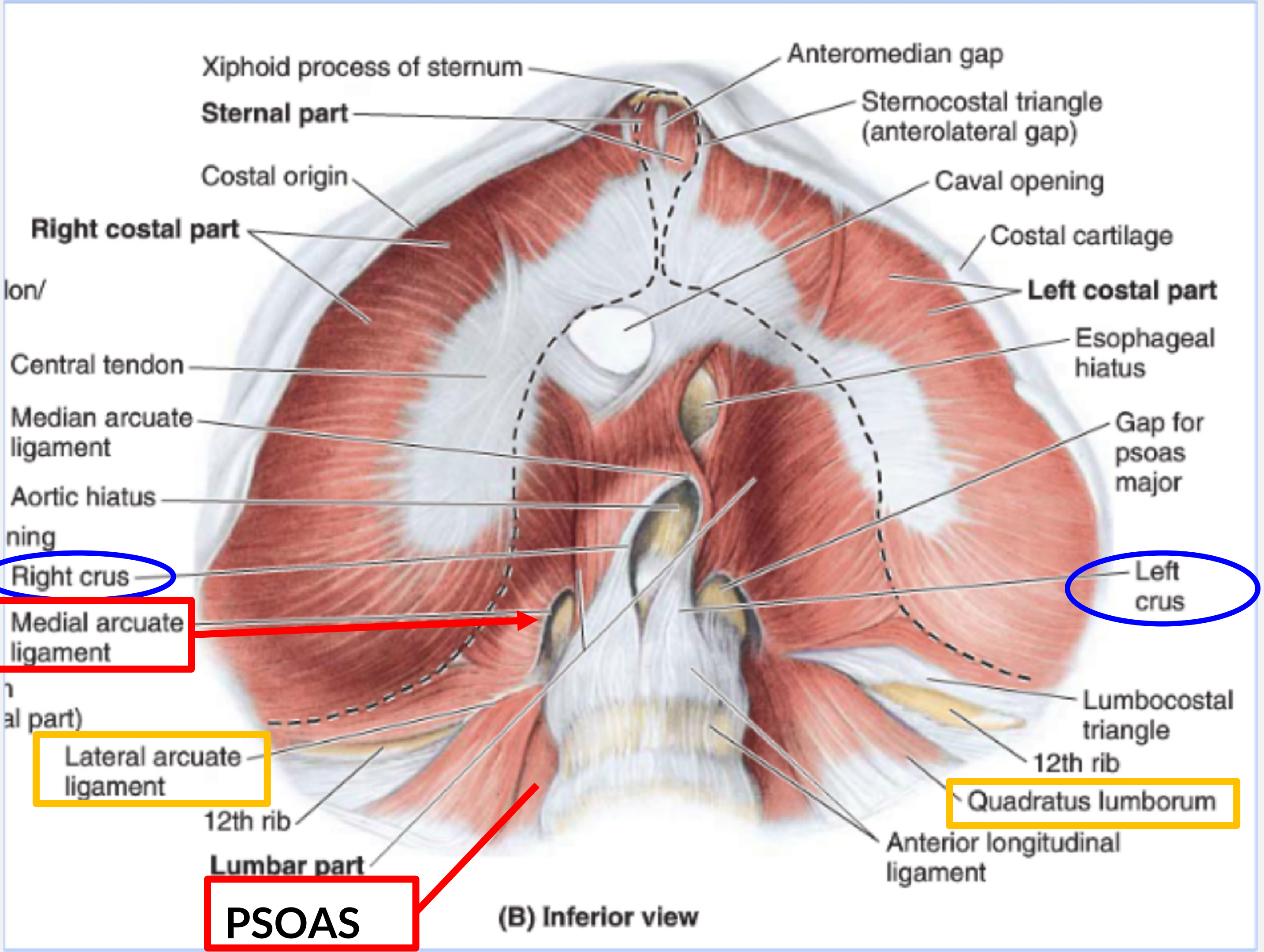
Drake: Gray's Anatomy for Students, 2nd Edition.


Right internal jugular vein
Brachiocephalic veins
Left internal jugular vein
Thoracic Duct
Collects lymph from the trunks labeled below
Left jugular trunk
Left subclavian trunk
Thoracic duct entering left subclavian vein
Left bronchomediastinal trunk
Thoracic duct
Thoracic lymph nodes
Parietal pleura (cut)
Diaphragm
cava (cut)
Cisterna chyli



M. longus capitis
s sympathicus cervicalis, R. communicans
superfic. (C 4)
R. communicans prof. (C 4, C 5)
Ganglion cervicale medium,
R. communicans C 5
Tuberculum caroticum, Angulus
scalenovertebralis
M. scalenus ant.
M. scalenus medius
A. V. vertebralis
N. vertebralis
V. cervicalis prof.
cervicale inf.,
communicantes
C 7, C 8, Th 1
tercostalis
suprema
alenus
post
ula
Sulcus
laviae,
bclavia
Pleura costalis
N. phrenicus
N. laryngeus sup. (R. ext.)
N. laryngeus sup. (R. int.)
A. laryngea sup.
A. thyroidea sup. (R. ant. et post.)
A. carotis communis
A. thyroidea sup. (R. cricothyroide
V. jugularis int., N. phrenicus
N. vagus, R. cardiacus sup.
A. thyr. sup. (R. marginalis su
V. isthmi
A. thyroidea inf.
A., V. vertebralis
A. cervicalis superfi
Truncus lymph. jugu
A. transversa col
A. suprascapul
Ductus thora
Angulus veno
A. sub
V. sub
Truncu
subclav
A. subclavia,
Ansa subclavia,
V. brachiocephal

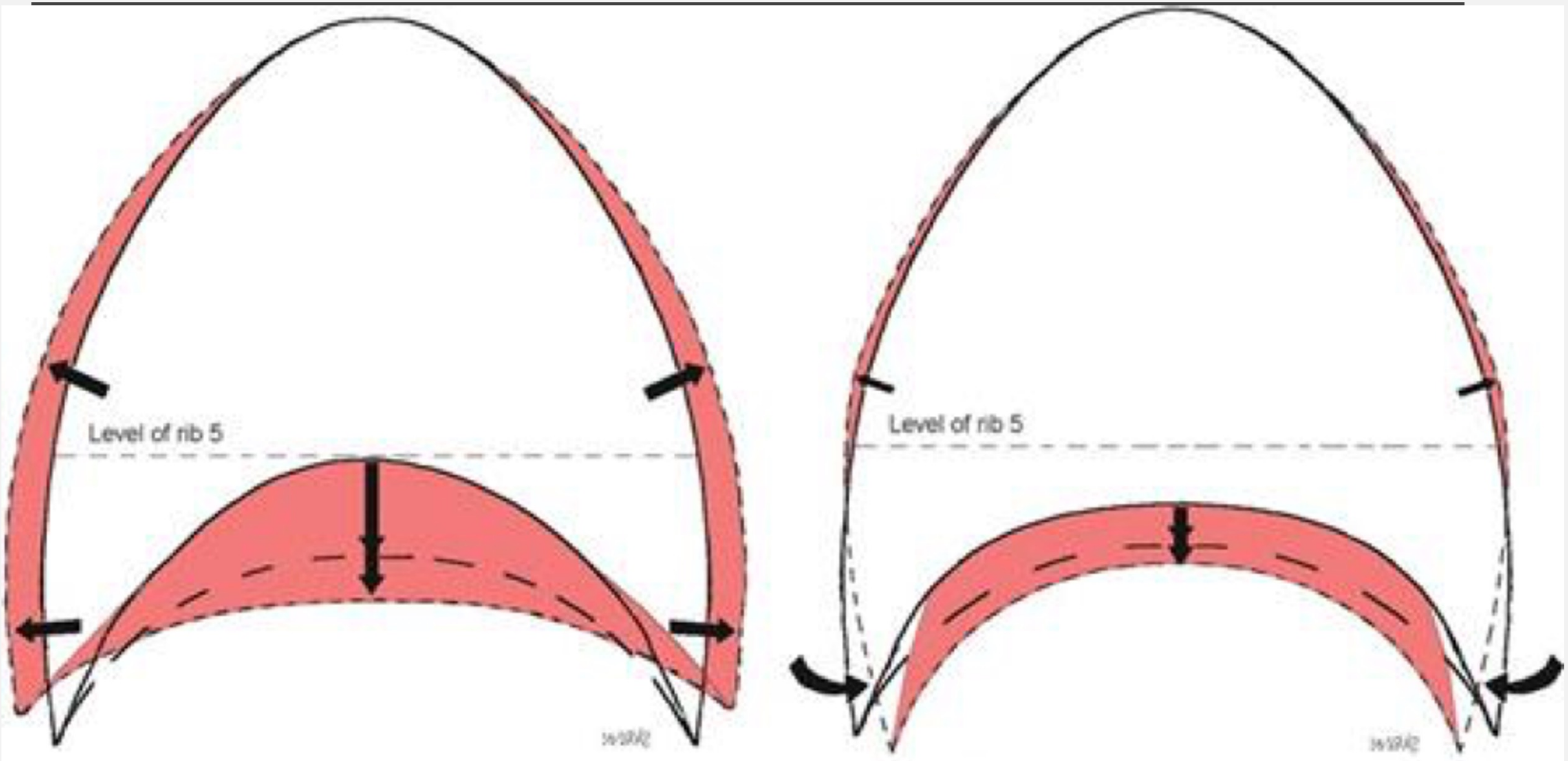

Suprapleural membrane
(Sibson's fascia)



(B) Inferior view

RESPIRATION

- Responsible for not only the movement of air, but also **venous and lymphatic circulation**
- Inferior vena cava and portal return are influenced by respiration, ***especially in the supine, non-active patient***
- **35-60%** of the total thoracic duct drainage is in response to movements occurring during respiration



Well-domed diaphragm with good contraction and good compliance of rib cage during inhalation

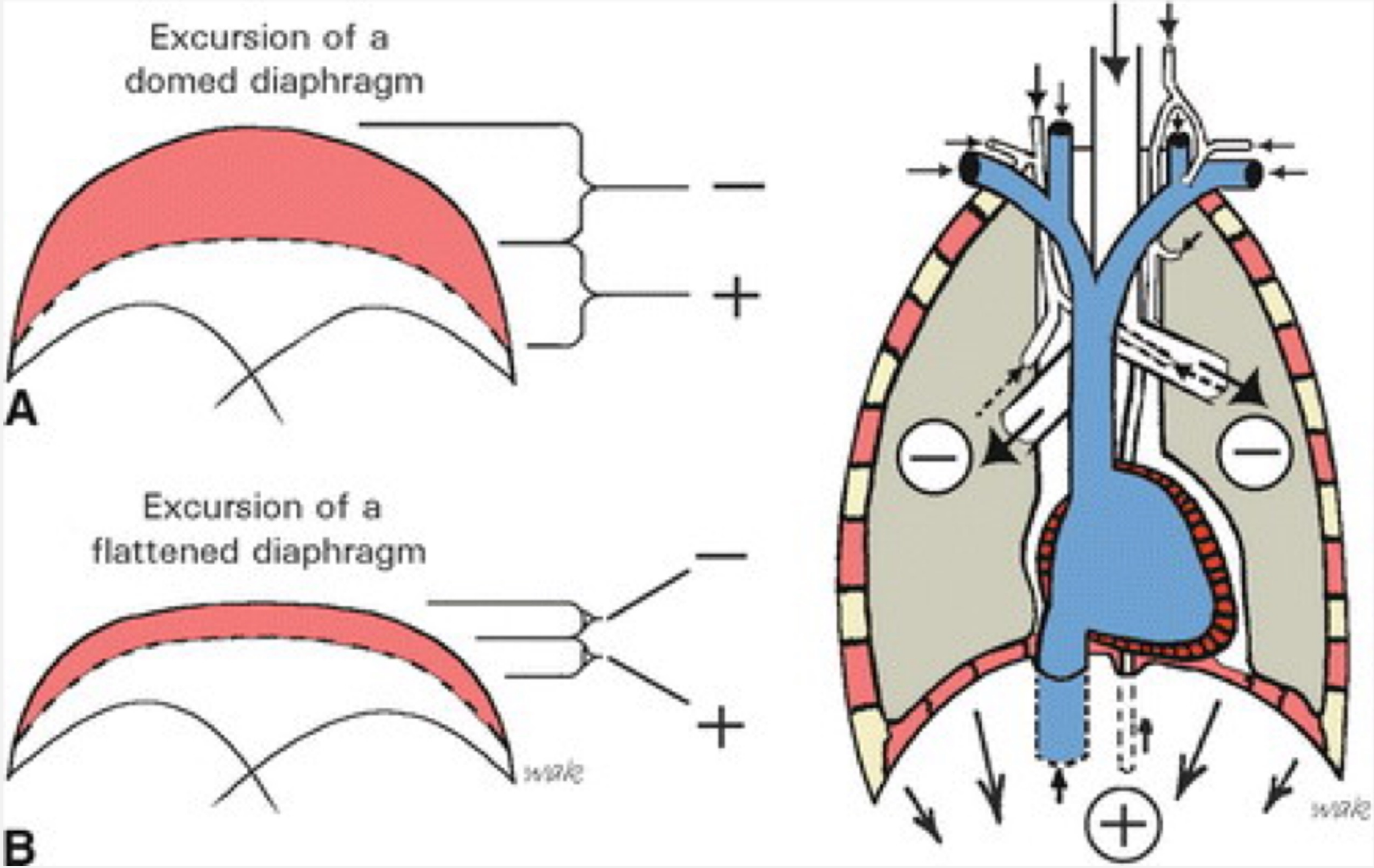
With contraction of well-domed diaphragm
With added compliance of thoracic cage

A

Flattened, spastic diaphragm, poor compliance and inward paradoxical movement of rib cage with inhalation

With contraction of flattened diphargm
Inward movement of thoracic cage

B


Excursion of a
domed diaphragm
−
+
A
Excursion of a
flattened diaphragm
−
+
B

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- Viscerosomatic reflexes

CASE 2:

- HPI: 7 yo male with PMH pneumonia w/ failed OP treatment 2/2 intolerance of prescribed antibiotics, presents to the ED with 10-day h/o worsening cough associated with yellow sputum and pain, admitted under impression of pneumonia. No known exposures, admission COVID test neg. ONMM service consulted to support patient's breathing and healing process.
- CC: pain in his chest with deep breaths, worsened by coughing, as well as decreased appetite, decreased UOP.
- ROS: + nausea w/ one episode of vomiting last night. Negative except as outlined above.
- Meds: Ceftriaxone, IV D5 NS

PHYSICAL EXAM (PE):

- Vitals: Temp **101.1**; HR **130**; RR **22;** BP: 90/70 O2 Sat: 96% RA
- Gen: resting, mild respiratory distress
- HEENT: normocephalic/atraumatic (NCAT); MM dry
- CV: tachycardic, regular rhythm, +S1/S2, no murmur, rubs, gallops, thrills
- Resp: **poor air movement throughout, left lower lobe with inspiratory rhonchi,** no wheeze, crackles.
- Abd: normoactive, soft, non-tender (NT), non-distended (ND)

OSTEOPATHIC STRUCTURAL EXAM (OSE):

- Head: OA ESRRL; decreased PRM
- Cervical: C2 RR
- Thoracic: CTJ FRSR, **T2-6 N SRRL w/ boggy paraspinals L>R**
- Lumbar: **TLJ flattened with boggy paraspinals L>R, decreased motion w/ respiration**; L1-L2 ERSL; L5 F RLSR
- Sacrum: base posterior; L SI compression
- Ribs: Left first rib inhaled, **left lower ribs exhaled**, **diffusely decreased compliance and excursion;** sternum decreased motion w/ respiration
- UE: **supraclavicular bogginess**
- Abdomen: **L>R hemidiaphragm inhaled; b/l crus restriction; L psoas hypertonic**

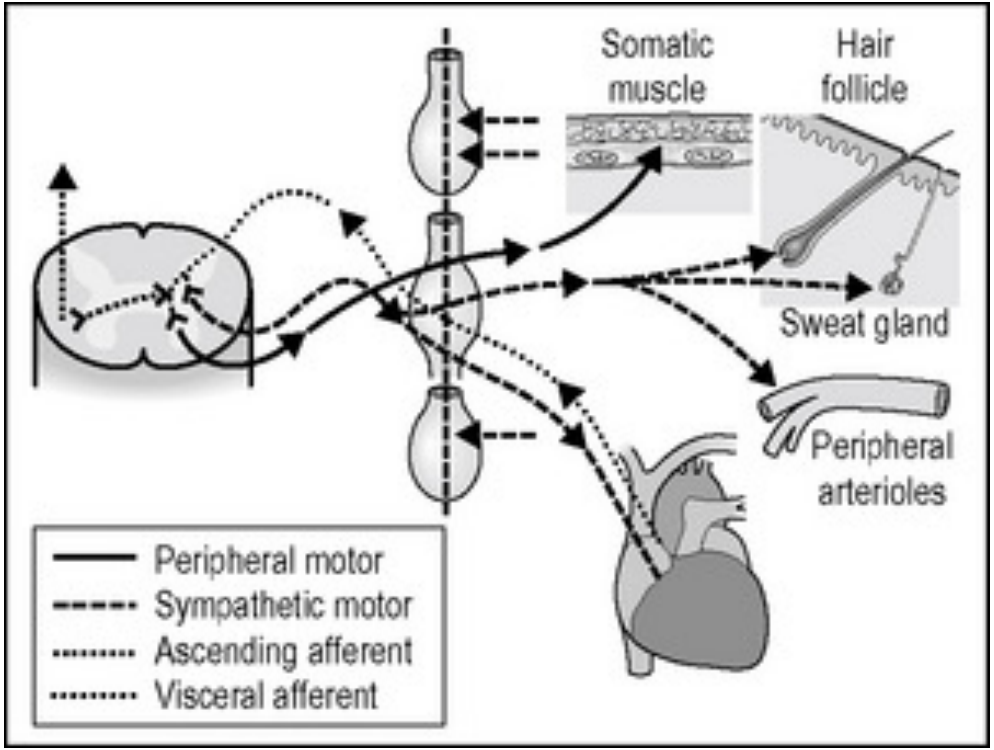
ASSESSMENT AND PLAN:

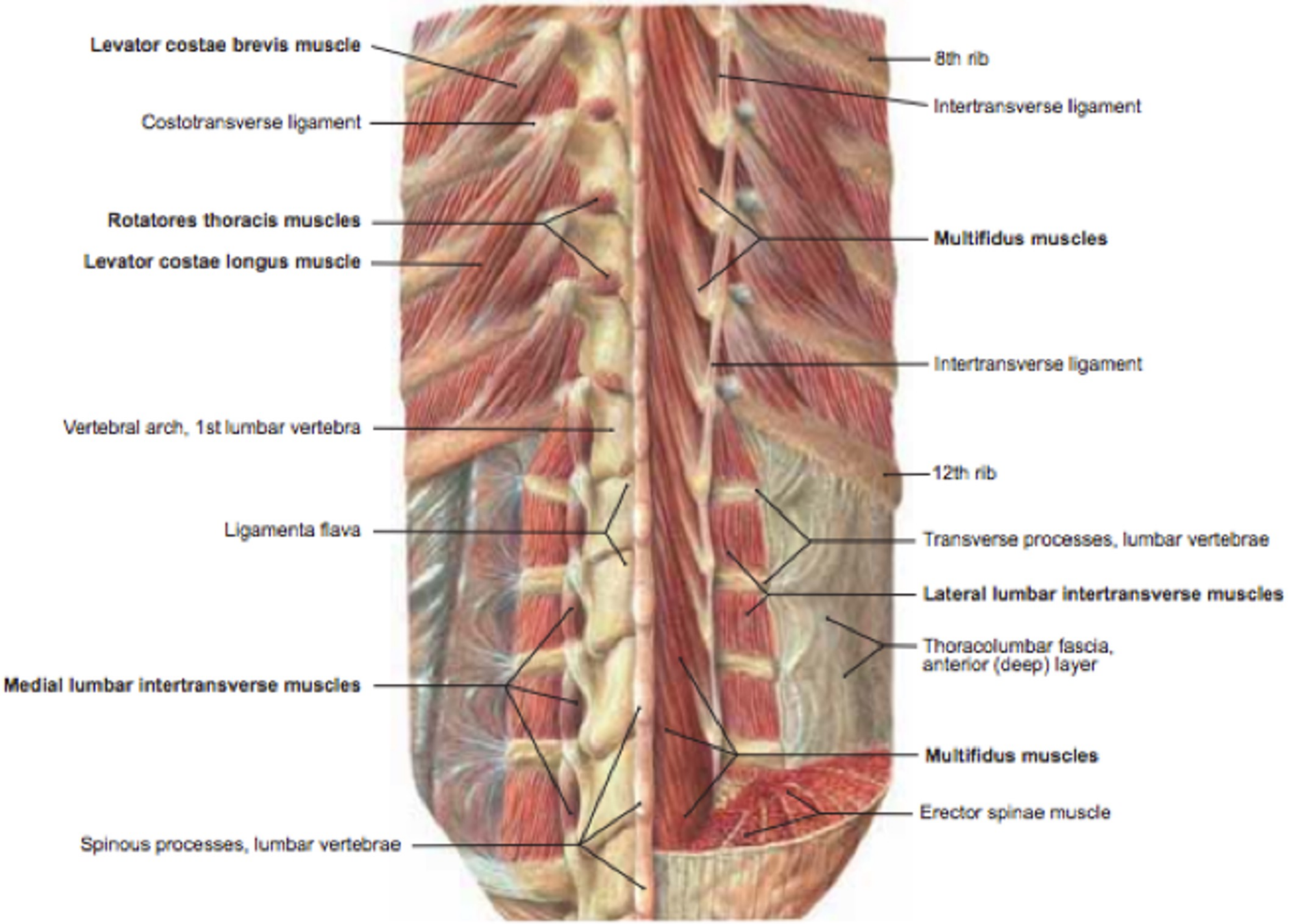
- 7 yo male w/ PMH PNA w/ failed OP treatment w/ LLL pneumonia, with pain on respiration. s/p IV Abx and fluids.
- Febrile, tachycardic, tachypneic, poor air mvt, rhonchi worst in LLL.
- OSE w/ above noted severe acute SD, most notably inhaled left hemidiaphragm, exhaled L lower ribs, congestion of thoracic inlet, and segmental facilitation of L T-spine c/w viscero-somatic reflex pattern, and respiratory-circulatory dysfunction. After discussing risks/benefits/alternatives, consent was obtained, and OMT, primarily BLT, was performed to palliate pain, decrease facilitation, and optimize resp-circ function to support lymphatic and immune function in LLL PNA. Treatment was well tolerated with improvement in severity of SD noted.

PNEUMONIA OF LLL

- Viscerosomatic reflex 2/2 inflammation of the lung
- Lymphatics
- Respiratory-Circulatory Dysfunction
  - Breathing
    - Rib dysfunction
    - Diaphragm SD

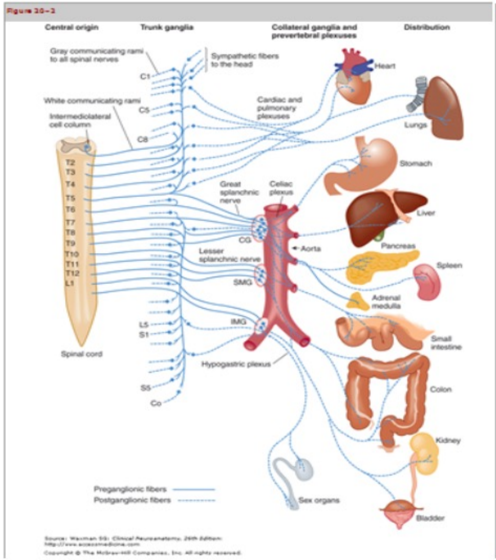
VISCERO-SOMATIC REFLEXES

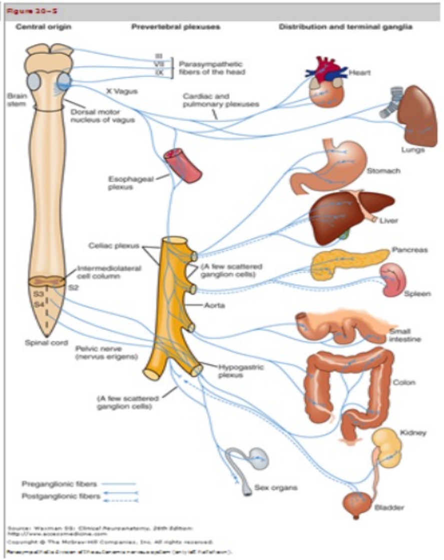




Levator costae brevis muscle
Costotransverse ligament
Rotatores thoracis muscles
Levator costae longus muscle
Vertebral arch, 1st lumbar vertebra
Ligamenta flava
Medial lumbar intertransverse muscles
Spinous processes, lumbar vertebrae
8th rib
Intertransverse ligament
Multifidus muscles
Intertransverse ligament
12th rib
Transverse processes, lumbar vertebrae
Lateral lumbar intertransverse muscles
Thoracolumbar fascia,
anterior (deep) layer
Multifidus muscles
Erector spinae muscle

VISCEROSOMATIC REFLEXES





Organ	Sympathetic Innervation	Parasympathetic Innervation
Head and Neck	T1- T5	Vagus (also CNIII, VII, IX)
Heart	T1- T5	Vagus
Lung	T2-T6	Vagus
Lower Esophagus / Stomach	T5-T10	Vagus
Liver and Gallbladder	T6-T9	Vagus
Small Intestine	T10-T11	Vagus
Ascending / Transverse Colon	T12-L1	Vagus
Descending and Sigmoid Colon / Rectum	L1-L2	S2-S4
Kidney	T10-L1	Vagus
Ureter	T10-L2	Vagus (proximal 2/3), S2-S4 (distal 1/3)
Ovary/Testes	T9-T11	S2-S4
Bladder	T10-L2	S2-S4
Uterus	T12-L1	S2-S4
Cervix	T10-L2	S2-S4

OSTEOPATHIC TREATMENT PLAN

- Developed as an extension of your focused structural exam
- **Individualized** for each acutely ill hospitalized patient **at each visit**
- Monitor effects of chosen technique while it is being performed (AS ALWAYS) and modify the technique as it is being applied

Sample treatment sequence:

1. Treat the viscerosomatic reflex with inhibition and/or by improving thoracic compliance and rib raising
2. Treat specific rib dysfunctions
3. Treat the thoracic inlet
4. Treat the thoracic diaphragm
5. Consider treating abnormalities to the Zink screen: L/S junction, T/L junction, C/T junction, OA
6. Treat other somatic dysfunction

RELATIVE CONTRAINDICATIONS

- HVLA & ME in sick patients
- Direct MFR near recent incisions
- Manipulation in area of known or suspected thrombus, DVT
- Manipulation to the cranium in a patient with stroke or head trauma, consider application of OCMM indirectly to sacrum
- Cancer (theoretical)--controversial topic- safest to avoid direct lymphatic pumps
- Counterstrain- be careful with positioning (Osteoporosis, lytic lesions, etc.)
- Do not manipulate the viscera if the tissue is inflamed
- Fractures
- Bacterial infection w/ temp >102° F

OPTIMIZE ORGAN SYSTEM FUNCTION TO IMPROVE CHANCES OF SURVIVAL/ PROMOTE HEALING AND HOMEOSTASIS

- Decrease length of stay
- Decrease exposure to pathogens
- Save $$$
- Avoid re-admission
- Improved immunity, oxygenation, perfusion
- Patient satisfaction/comfort
- Prevent/treat common complications of hospitalization and surgery: atelectasis/pneumonia, adynamic ileus, deep vein thrombosis (DVT), poor wound healing

BACK TO OUR OSTEOPATHIC PRINCIPLES:

- Rational treatment is based upon an understanding of the basic principles of body unity, self-regulation and the interrelationship of structure and function.

- *"We say disease when we should say effect; for disease is the effect of a change in the parts of the physical body. Disease in an abnormal body is just as natural as is health when all the parts are in place." - A.T. Still*

OMM FOR THE SURGICAL PATIENT

STACIA SLOANE, DO

Previous Iteration by Melissa G. Pearce, D.O., C.S.

SURGICAL OUTCOMES

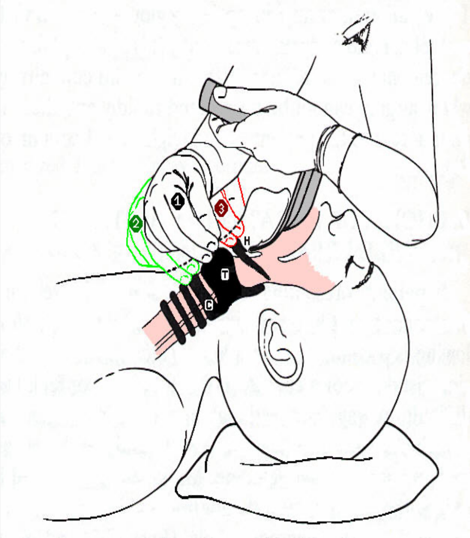
- Goal to avoid post-procedure adverse events
  - negative effect on patient health
  - negative financial outcomes for hospitals related to adverse events
- Strategies to avoid
  - atelectasis /pneumonia
  - Ileus
  - deep vein thrombosis
  - wound infection
  - hospital readmission

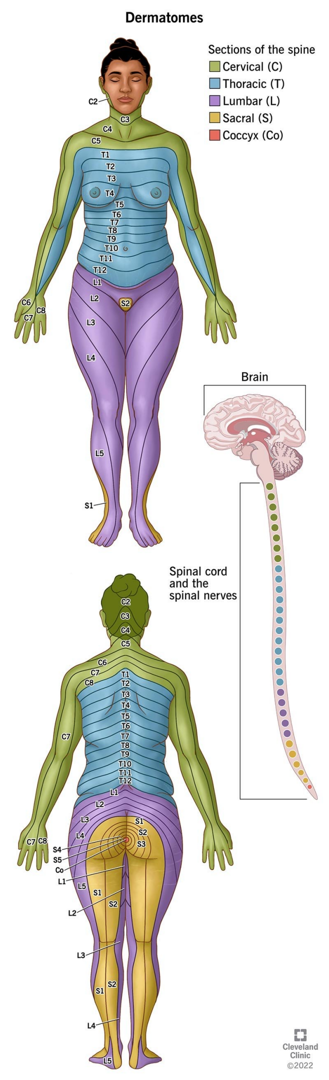
Scientificamerican.com

ATELECTASIS/PNEUMONIA

Risk factors

- Hyperextension for intubation
  - somatic dysfunction at C3-5
  - negative effect on phrenic nerve
- Surgical incision, abdominal
  - disrupts dermatomes for T7-T12
  - negative effects on intercostal nerve function





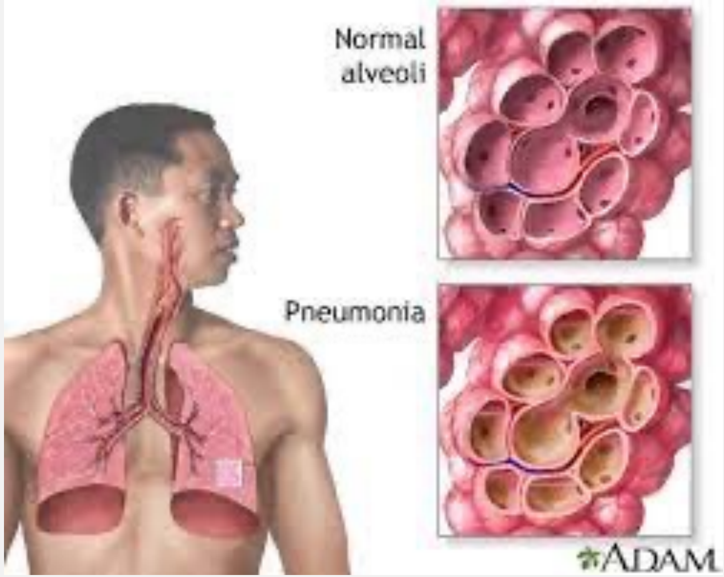
ATELECTASIS/PNEUMONIA

Post-procedure pneumonia

- 3rd most frequent hospital-acquired infection
- 15% of infections

Common progression

- restricted breathing patterns post-operatively
- atelectasis
- pneumonia



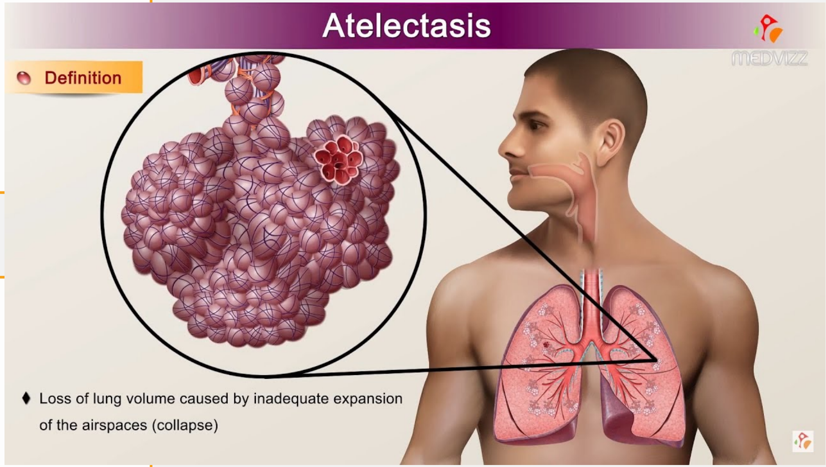
ATELECTASIS/PNEUMONIA

Conventional prevention

- early movement
- bedside spirometry
- adequate pain control

Osteopathic intervention – treat

- viscerosomatic reflex(es), normalize sympathetic/parasympathetic balance
- diaphragms, optimize lymphatic function
- somatic dysfunction in the thoracic region



ATELECTASIS/PNEUMONIA STUDIES UTILIZING OMT

Atelectasis

1) Sleszynski, et. al

- Compared bedside spirometry w/twice-daily OMT
- Similar outcomes, fewer interventions needed w/OMT, earlier return to baseline PFTs

Pneumonia

2) Noll, et. Al

58 patients hospitalized w/pneumonia, single blind, randomized study

End-points: number of days of antibiotic use, number of hospital days

Statistically significant decrease from 8.1 to 6.1 total antibiotics and 8.5 to 6.5 hospital days

POST-OPERATIVE ILEUS

- Impairment of intestinal motility
  - produces discomfort
  - may prolong hospital stay
  - incidence 8.5% after abdominal surgery (range 4.1-19.2%)
- Treatment - routine
  - nasogastric tube placement
    - may delay return of motility
  - manual dis-impaction

POST-OPERATIVE ILEUS

- Treatment - novel
  - selective mu-receptor antagonistic
    - alvimopan
    - https://emedicine.medscape.com/article/2242141-medication#2
  - chewing gum
- OMT to reduce progression from slowed motility to ileus and restore function

POST-OPERATIVE RETURN OF BOWEL FUNCTION AND LENGTH OF STAY

Baltazar et. al. http://jaoa.org/article.aspx?articleid=2094531

- Retrospective chart review
- 55 patients status post major GI surgery
  - 17 received OMT, 38 did not receive OMT
- Evaluation for:
  - time to flatus
  - time to clear liquid diet
  - time to bowel movement
  - length of stay
- Standardization for multiple factors, some of which are:
  - anesthesia classification
  - comorbid conditions
  - complications
  - type of narcotics used

POST-OPERATIVE RETURN OF BOWEL FUNCTION AND LENGTH OF STAY

Bowel function

- Time to flatus
  - Non-OMT group 4.7 days
  - OMT group 3.1 days
  - Difference 1.6 days
- Time to bowel movement and clear liquid diet
  - No significant difference

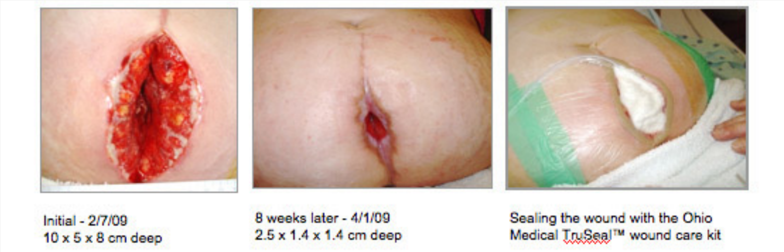
Length of stay

- Non-OMT group11.5 days
- OMT group 6.1 days
- Difference 5.4 days
- Expected to produce significant cost savings

SURGICAL SITE INFECTIONS

- Rate of post-operative surgical site infections
  - 2-7% depending on classification of clean or dirty wound
- Important cause of post-operative morbidity and mortality
- Potential progression
  - cellulitis
  - abscess
  - advancing infection
  - sepsis

Wound Dehiscence



moblvacnpwt.com

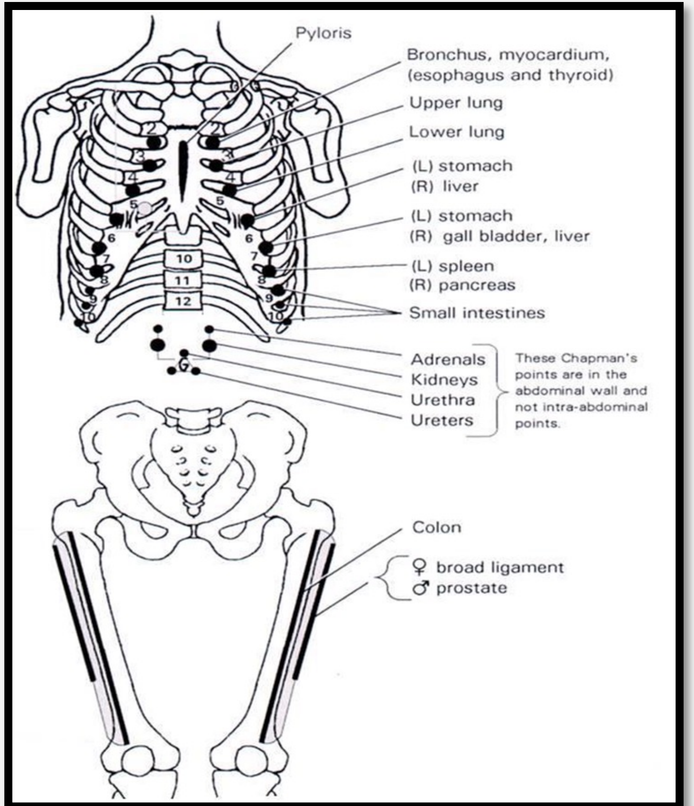
POST-SURGICAL PATIENT EXAMPLE

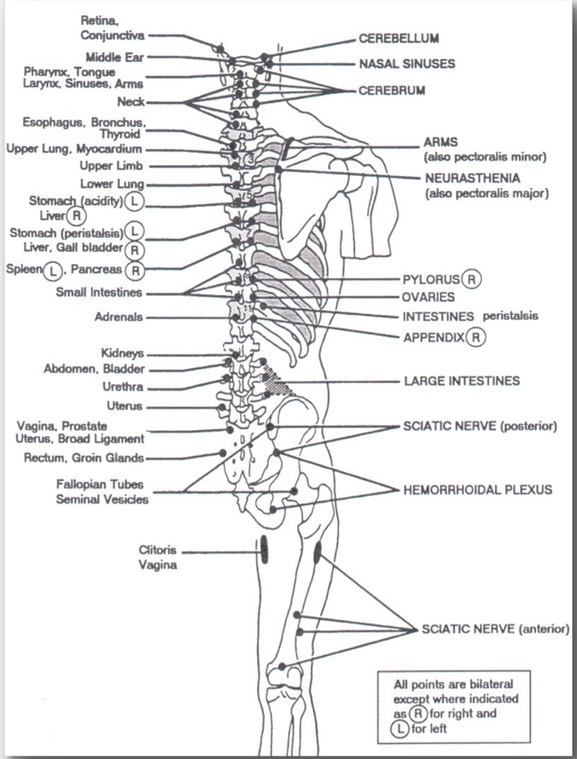
- Your patient with diverticulitis with resection of necrotic bowel and reanastomosis has had a good response to IV antibiotics and has changed to PO after three days in anticipation of discharge
- Pain reduced, fever resolved
- Treatment course included daily OMT with good response to somatic dysfunction findings
- On day of planned discharge, patient spiked a temp to 100.6 degrees F
- What issues are you considering?

POST-SURGICAL PATIENT EXAMPLE

- Patient reports shortness of breath on ambulation, no chest pain, no leg pain
  - VS T 98.7, BP 126/77, P 96, R 28, O2 sat 96%
  - RRR no M/G/R
  - Lungs with coarse breath sounds at B bases
  - Abd soft, mildly tender in LLQ, no rebound
  - Decreased diaphragmatic excursion
  - Normal pedal pulses, no calf or pedal edema
  - Warm, boggy tissues T4-7
  - Exquisitely tender area in bilateral 4th intercostal spaces
- What are your next steps?

CHAPMAN'S POINTS





POST-SURGICAL PATIENT EXAMPLE

- Chest Xray with bibasilar lung opacification, no mediastinal shift, no consolidations
- What are your next steps?

- Incentive spirometry
- OMT
  - thoracic paraspinal inhibition
  - diaphragmatic release
  - rib-raising

HOSPITAL LENGTH OF STAY

Health Catalyst report on El Camino Hospital operational changes to address hospital length of stay (LOS)

- Shorter length of stay results in:
  - reduction in readmission rates
  - decrease in Hospital Acquired Conditions (HAC)
  - decreased incidence of rate of patient safety indicators
  - multi-million dollar projected annual cost savings

POST-SURGICAL RESEARCH FINDINGS EFFECTS ON LENGTH OF HOSPITAL STAY

- Retrospective Chart Review
- Computer search screen
  - ICD-9 for Ileus (n=655)
- Manual Chart review
  - n=331 had abdominal surgery
  - Two groups
    - Received OMT n=172
    - Did not receive OMT n=139

Crow, *International Journal of Osteopathic Medicine 2009*

POST-SURGICAL RESEARCH FINDINGS EFFECTS ON LENGTH OF HOSPITAL STAY

- OMT consults to treat post-op ileus by surgeon request
- OMT was done by
  - 10 faculty (9 DOs and 1 MD)
  - Osteopathic medical students
  - Osteopathic family practice residents
- No predetermined protocol or approach for evaluation or treatment

Crow, *International Journal of Osteopathic Medicine 2009*

POST-SURGICAL RESEARCH FINDINGS EFFECTS ON LENGTH OF HOSPITAL STAY

- Group comparison
  - No statistical difference - gender, race/ethnicity, medication
  - Statistical difference - age
    - OMT vs no OMT group 53.6 vs 62.8 yrs old
- Outcomes - Mean Length of Stay (LOS)
  - OMT group = 11.8 days
  - non-OMT group =14.6 days
  - Difference = 2.7 days
  - Corrected for age difference = 4.1 days

Crow, *International Journal of Osteopathic Medicine* 2009

HOSPITAL LENGTH OF STAY

- The effect of osteopathic manipulative treatment on length of stay in posterolateral postthoracotomy patients: A retrospective case note study, 2015
  - Fleming, et. al, Kirksville, MO, retrospective review 1998 - 2011
  - 38 patients, 23 received OMT, 15 did not receive OMT
  - Hospital LOS 11 days versus 10.5 days, not statistically significant difference
    - All patients with post-op ileus or ICU admission were consulted for OMT
    - Severity of illness may have been greater in the treated patients

- A more generalized retrospective review in the same hospital setting evaluated 1509 patients seen by the OMM consult service and found LOS average 5.7 days without OMT and 3.1 day with OMT

OPTIMIZATION OF CARE FOR THE TOTAL ABDOMINAL HYSTERECTOMY (TAH) PATIENT

Goldstein, *JAOA 2005*

- Study Goal – To evaluate pre-op morphine and post-op OMT post-op pain
- Methods – 39 patients randomized to one of four groups:
- Pre-op saline & post-op sham OMT
- Pre-op saline & post-op OMT
- Pre-op morphine & post-op sham OMT
- Pre-op morphine & post-op OMT
- Results for groups with pre-op morphine
  - Post-Op OMT group vs Sham
    - Used less total morphine in first 48 hours post-op
- Conclusions:
  - OMT can be a useful adjunct in surgical pain management for these patients

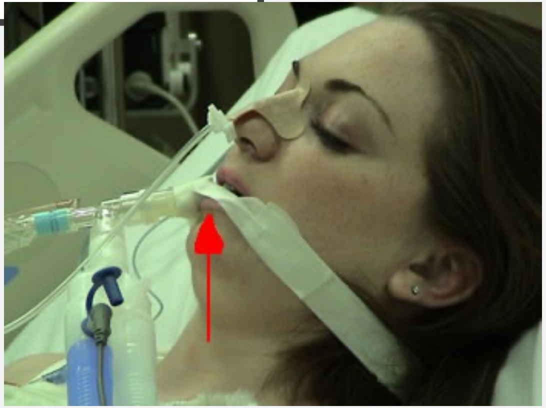
OPPORTUNITIES IN CLINICAL SETTING

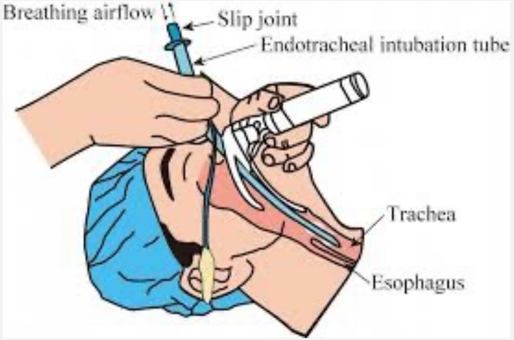
Postoperative complication reduction - preop

- Begin with preoperative clearance visit
  - Optimize lymphatic function, diaphragmatic motion
  - Reduce cervical region somatic dysfunction
  - Improve biomechanics of region involved in surgery
  - Minimize associated viscerosomatic, Chapmans reflexes
  - Treat visceral structures, as appropriate

Postoperative OMT

- Schedule postoperative visit within 1-2 weeks of surgery





OPTIMIZATION OF CARE FOR THE ORTHOPEDIC SURGICAL PATIENT

- Important things to consider in the post-operative orthopedic patient include appropriate application of orthopedic exam maneuvers, osteopathic structural exam, and treatment positions
  - Patients immediately after rotator cuff repair should not be moved into the physiologic barrier
  - Patients after total hip replacement (THR) must avoid hip flexion beyond 90 degrees, marked hip internal or external rotation, and marked abduction
    - What implication does this have for the application of OMT?

www.healthline.com

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